

Human-Calibrated Risk-Sensitive Decision Making for Non-Intrusive Assistive Mobile Manipulation

Water delivery as a concrete testbed for deciding whether to hand over directly, hand over cautiously, place the object nearby, wait safely, or retreat

Core contribution: This project reformulates service-oriented mobile manipulation from a pure “task completion” problem into a risk-sensitive service-decision problem, and builds an interpretable prototype evaluated through policy comparison, ablations, safety stress tests, preference calibration, and Webots/RViz branch-mapping validation.

1. Problem and research gap

Service robots should not only maximize direct task completion. In human-facing service tasks, direct handover may be intrusive, uncomfortable, or unsafe when external events or environmental risks appear. A robot needs to reason about **user state, external events, environmental risk, and individual preference**.

Water delivery is used as a concrete testbed; the broader target is non-intrusive assistive mobile manipulation.

2. Method and system structure

Service decisions are decomposed into three information sources: user state captures service intent and interruption sensitivity; external events indicate whether the robot should pause, yield, or re-evaluate; environmental risk indicates whether approaching, placing, or retreating is safe.

Inputs and actions

Inputs:

- User state: focused / requesting water / resting / available for interaction / uncertain
- External event: third-party passing / third-party intervention / path blocked / recognition uncertainty
- Environmental risk: near-user blockage / side-table blockage / retreat-path blockage / high dynamic risk

Actions: direct handover / cautious handover / nearby placement / safe waiting / retreat

Three-tier prototype

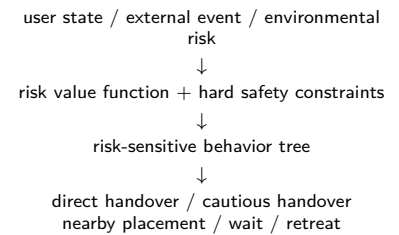
Deliberative layer: decision sandbox / risk-sensitive behavior tree selects the service action.

Sequencing layer: bridge maps the action to navigation goals, arm poses, waiting, or retreat branches.

Execution and visualization layer: Nav2/Webots handle path execution and simulation; RViz visualizes paths, markers, logs, and branch validation.

Decision logic: hard safety constraints first mask infeasible actions; the remaining action with minimum weighted risk cost is selected.

Decision pipeline



3. Evidence chain and key results

Evidence chain: policy comparison → ablation study → safety stress test → 35 direct preference labels → Webots/RViz branch mapping.

Policy	Action correctness	Unsafe continuation	Avg. risk cost
Fixed policy	0.5515	0.1636	55.39
Rule-based policy	0.8061	0	26.65
Risk-sensitive policy	0.9818	0	18.57

Note: the fixed policy tends to directly hand over; the rule-based policy follows hand-coded if-then rules; the risk-sensitive policy minimizes weighted risk cost under hard safety constraints. Ablation: removing external-event modeling increases unsafe continuation from 0.0704 to 0.2084.

Safety fail-safe

Tempting direct-handover stress test:
Without safety override: hard safety violation = **1.0**
Full policy: hard safety violation = **0**
When the cost model makes a dangerous action look attractive, hard safety still blocks it.

Preference calibration

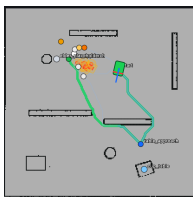
35 direct annotations: cautious handover = **11**, nearby placement = **10**, safe waiting = **9**, direct handover = **0**, retreat = **0**.
Service completion should not be defined only as direct handover; nearby placement can also be an appropriate service-completion mode.

4. Current scope → PhD direction

Current scope	PhD direction
State is represented by discrete labels	→ Continuous belief states for intent strength, attention, uncertainty, and path-risk confidence
Evaluation rules are still hand-designed	→ More preference feedback to learn or calibrate risk weights
Preference update is rule-based	→ Safety-constrained preference learning for personalized soft-risk boundaries
Webots/RViz validates branch mapping	→ Online observe–decide–act–reassess loop
Water delivery is the current testbed	→ Medication delivery, object handover, reminders, and other assistive services

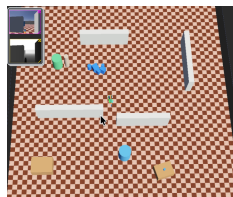
5. Demo evidence: decision sandbox → Webots execution → RViz validation

Decision sandbox



context → selected_action

Webots execution



selected service branch execution

RViz validation



path, marker, and log visualization